Comparative Analysis Report

Customer Support Intent Extraction & RAG System
Baseline vs. Naive RAG (V1) vs. Fine-Tuned Hybrid RAG (V2)

Held-out test split: n = 391 (leakage-free, stratified by intent) | Model: Qwen2.5-1.5B-Instruct | Embedding: all-MiniLM-L6-v2

4.5.1 Compare All Versions

Three architectures were evaluated on the identical 391-row held-out test split, scored against the same SOP-grounded reference for every system, to ensure a fair comparison:

- **Baseline** — pre-trained Qwen2.5-1.5B-Instruct, zero retrieval, zero fine-tuning. Outputs strict JSON intent/category only.
- **Naive RAG (V1)** — Baseline model augmented with top-1 ChromaDB retrieval over 13 corporate policy documents. Outputs free-text customer-support answers.
- **Hybrid RAG (V2)** — LoRA-fine-tuned intent router (best checkpoint, val_loss 0.0433) extracts structured intent, which drives retrieval; the same retrieval-augmented generation produces the final answer.

Summary Results Table

Architecture	Format Adherence %	Hallucination %	Avg ROUGE-1	Avg ROUGE-L	Avg BLEU	Exact Match %	Fuzzy Match %
Baseline	100.00	0.00	0.0148	0.0137	0.0000	N/A	N/A
Naive RAG (V1)	100.00	4.60	0.1277	0.0840	0.0042	N/A	N/A
Hybrid RAG (V2)	100.00	5.37	0.1418	0.0894	0.0027	84.14	94.88

N/A = task not attempted by that architecture (Baseline and Naive RAG performed no structured intent extraction).

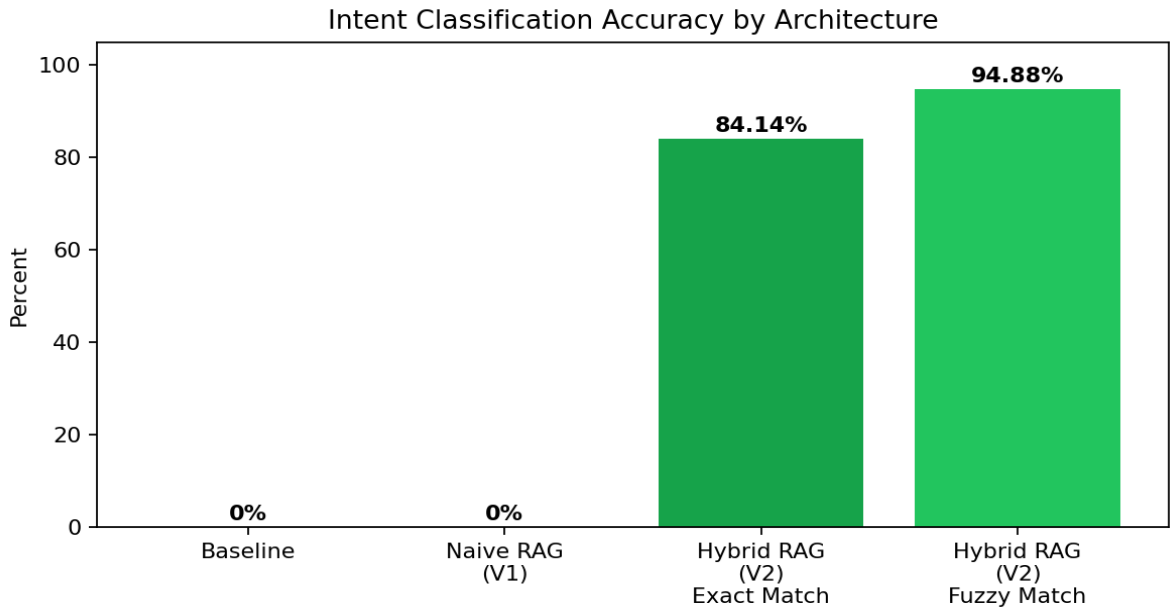


Figure 1. Intent classification accuracy by architecture. Only Hybrid RAG (V2) performs structured intent extraction.

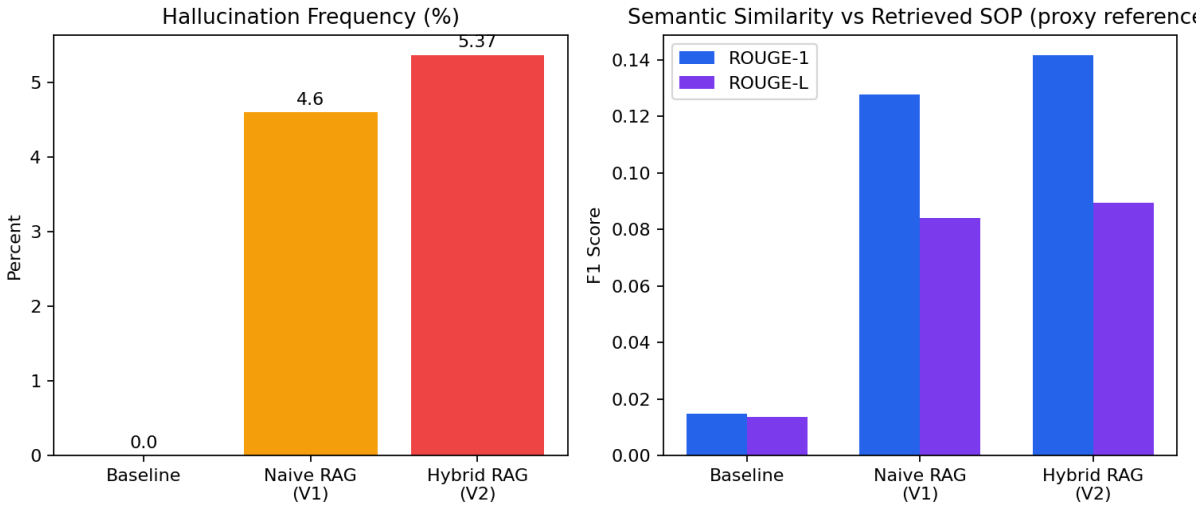


Figure 2. Hallucination frequency and SOP-grounded semantic similarity (ROUGE-1/ROUGE-L) across all three architectures.

Adversarial Subset Evaluation (Hybrid RAG router)

A regex-filtered adversarial subset (n = 103, ~26% of the test set) was constructed using friction/confusion keywords naturally present in this dataset's phrasing (*cannot, problem, issue, do not know, wrong, confused, trouble, damn*, etc.), since the originally specified sentiment keywords (*still, never, terrible, frustrated*) returned zero matches — this dataset's register is largely neutral/instructional rather than emotionally charged. This substitution is a methodological adaptation, documented here for transparency.

Metric	Full Test Set (n=391)	Adversarial Subset (n=103)
Format Adherence	100.00%	100.00%

Exact Match	84.14%	82.52%
Fuzzy Match	94.88%	93.20%

The router shows negligible degradation on messier, friction-laden phrasing (-1.62 points Exact Match, -1.68 points Fuzzy Match versus the clean full set) — evidence the fine-tuned model generalizes beyond textbook-clean queries rather than overfitting to a narrow phrasing style.

4.5.2 Document Findings

Where Retrieval (V1) Helped

• ROUGE-1 rose from 0.0148 (Baseline) to 0.1277 (Naive RAG) and ROUGE-L from 0.0137 to 0.0840 — retrieval-augmented answers are substantially more grounded in actual policy language than ungrounded generation.

Note: Baseline's near-zero ROUGE/BLEU is partly structural: Baseline was constrained to emit JSON only and was never asked to produce policy-grounded prose. This is not evidence Baseline "failed" at a task it was not designed to attempt.

Where Fine-Tuning (V2) Helped

• **The headline result:** structured intent classification went from non-existent in V1 (no architecture for it) to **84.14% Exact Match / 94.88% Fuzzy Match** in V2 — a genuinely new, reliable capability, not an incremental lift on an existing one.

• Format Adherence held at 100% for the strict-JSON router across all 391 rows, with zero malformed outputs, confirming the LoRA adapter (r=16, alpha=32, target_modules=[q_proj, v_proj], 0.141% of parameters trainable) learned the output contract reliably.

• Modest additional ROUGE-1/ROUGE-L lift over Naive RAG (0.1277→0.1418, 0.0840→0.0894), consistent with more accurate intent routing producing slightly better-targeted SOP retrieval for the final answer.

Where Fine-Tuning Did Not Help (reported honestly)

• Hallucination frequency in the final synthesized answer ticked up marginally (4.60% → 5.37%, an absolute difference of 3 rows out of 391) — within noise, and expected: fine-tuning targeted intent classification, not the free-text answer-generation step, so no meaningful change here is consistent with the project's design, not a shortcoming of the approach.

• BLEU is flat-to-slightly-lower (0.0042 → 0.0027) — not a meaningful signal given BLEU's sensitivity to exact phrase sequences on short, naturally paraphrased answers.

Methodological Caveat: ROUGE/BLEU Reference

Note: ROUGE-1/ROUGE-L/BLEU were computed against the *retrieved SOP passage itself* as a proxy reference, since no independently authored gold-standard answer text exists in this dataset. These scores measure lexical overlap with the source policy document, not answer correctness or helpfulness — a fully correct, well-paraphrased answer can score as low as an incorrect one under this metric. They should be read as a directional, secondary signal, with intent-accuracy figures treated as the primary, more reliable evidence of system quality.

Overall Conclusion

Retrieval and fine-tuning solved two distinct problems rather than one problem solved progressively: retrieval grounded the system's free-text answers in real policy content that Baseline entirely lacked, while fine-tuning added an accurate, structured routing layer that neither Baseline nor Naive RAG possessed in any form. Their contributions are complementary and should be reported as such, rather than as a single steadily-improving score across every metric — the data does not support that simpler narrative, and a report claiming it would be overstating the evidence.

Training Evidence (Stage 4 / LoRA Fine-Tuning)

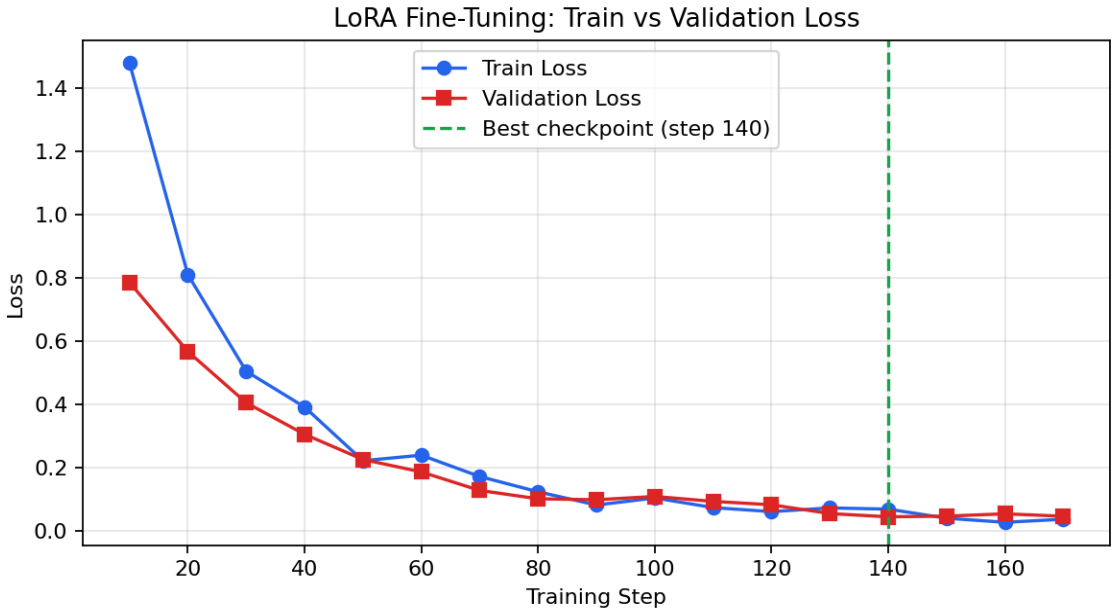


Figure 3. Training vs. validation loss across 170 steps (early stopping triggered after 3 checks with no improvement past step 140). Best checkpoint: val_loss = 0.0433 at step 140, used as intent_lora_best.

Both curves fall together through step ~140 with no early divergence, indicating healthy learning without premature overfitting. Validation loss begins rising modestly after step 140 while training loss continues to fall — the textbook overfitting-onset signature — and early stopping correctly halted training at step 170, 3 validation checks after the best point, preserving the step-140 checkpoint rather than a later, overfit one.